THE SMAKS PROJECT · KOCHI



Moonlight

Old medicines, new jobs. An AI explorer for drug-repurposing clues, built on open data.

Working engine and web page, built by a 9-year-old tech lead

Startup Buildathon India 2026 · Google DeepMind × Google Cloud

01 · THE PROBLEM

The evidence for repurposing old medicines already exists. Almost nobody can read it.

Repurposing is the fast lane

An approved medicine has passed safety trials. A new use can be studied in a fraction of the time and cost of inventing a new drug.

Rare diseases need it most

Thousands of rare diseases get little research funding. Patient groups hunt for leads with no bioinformatics staff.

The data is locked

Open Targets links drugs, proteins and diseases, but behind GraphQL, ontology IDs and dense score tables.

02 · THE IDEA

A medicine built for one disease may grab a protein that belongs to another.



Medicine

approved, already tested in people

Protein

the knob the medicine turns up or down

Disease

linked to that protein in the research record

The technique is called knowledge-graph drug repurposing. Moonlight finds the clues; scientists test them.

03 · HOW IT WORKS

Five stages. AI at both ends, verified data in the middle.

1 · TRANSLATE

Gemini

Everyday words like “flu”, misspellings or “high blood pressure” become disease names. Open Targets confirms the real ID.

IN FINAL REVIEW

2 · FETCH

Python + GraphQL

Disease to associated targets with scores; targets to drugs with mechanism, action type and phase.

3 · FILTER + RANK

Our graph engine

Drop drugs already linked to the disease. Rank the rest by proteins hit, weighted by association score.

4 · EXPLAIN

Gemini

Each path’s records become plain language. Input is only the retrieved data. No invented facts.

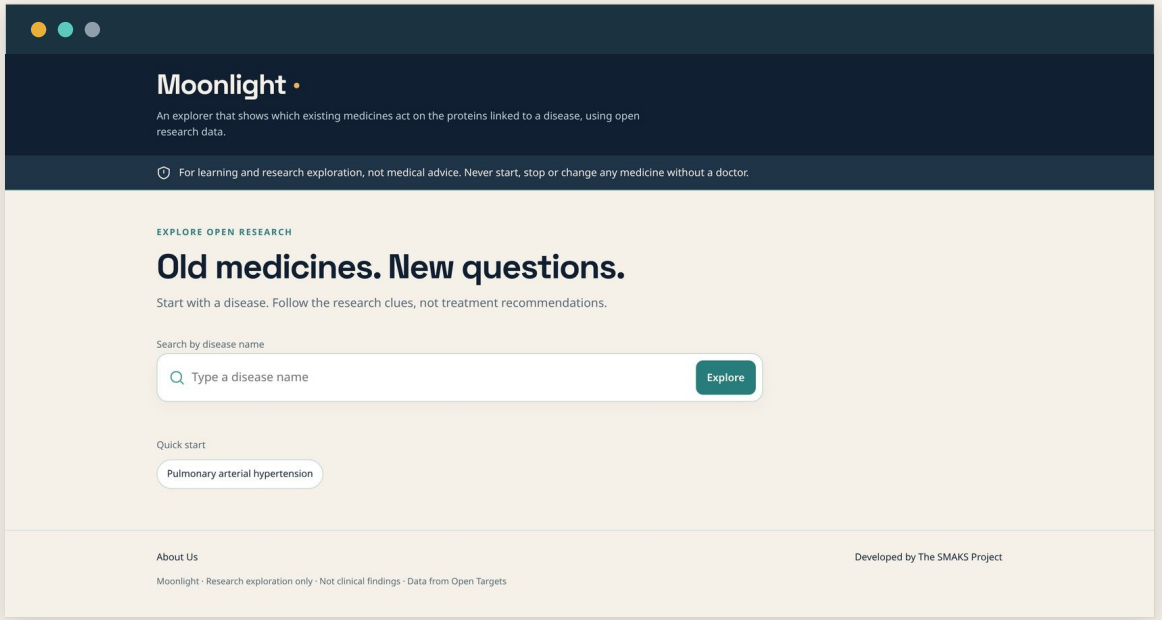
5 · WEB PAGE

FastAPI + HTML

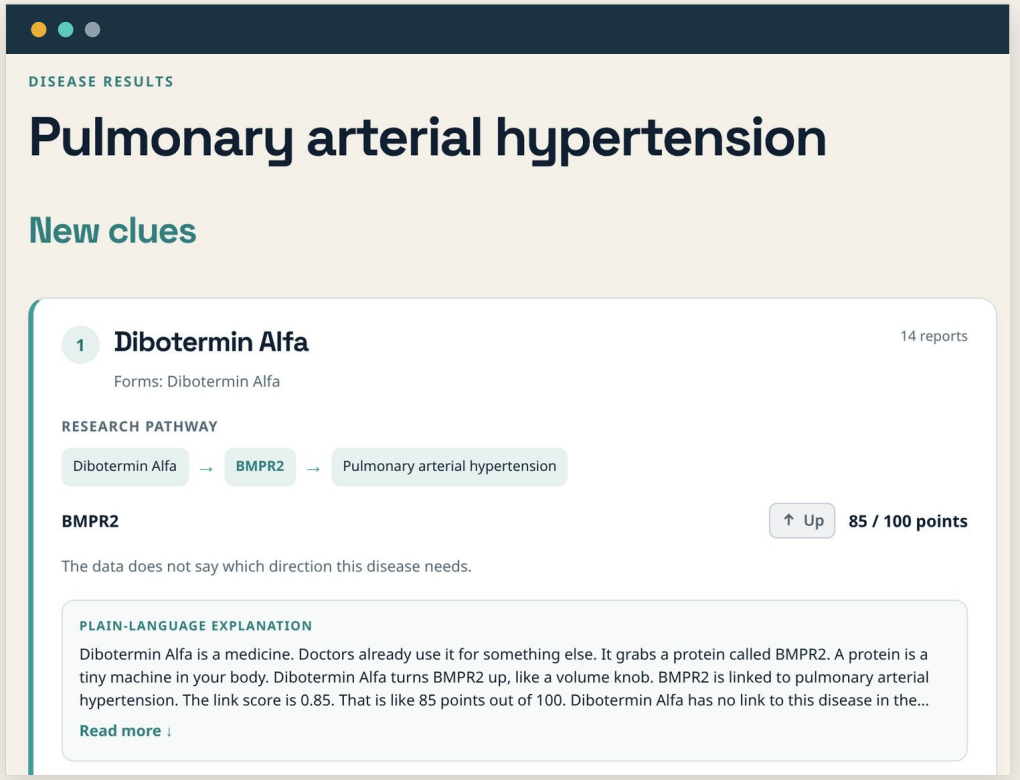
Designed by Sidha Mogha (Cochin Design Company). Medicines first, then the protein path, with source and evidence strength.

Dark stages are AI. Light stages are deterministic code and verified data.

A working page, built on live open data.



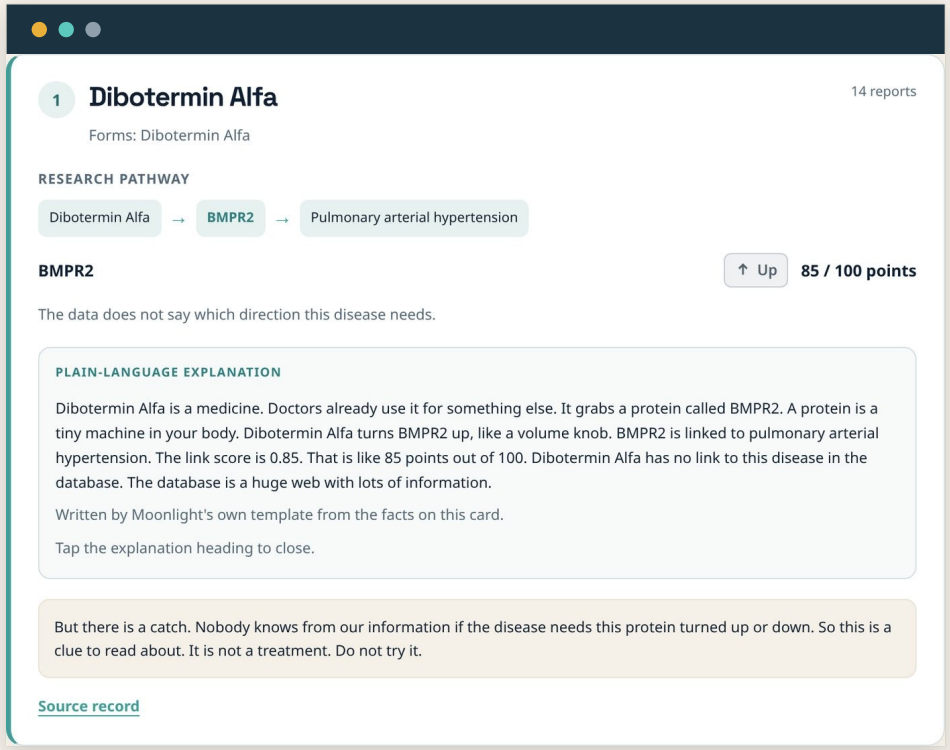
Search a disease



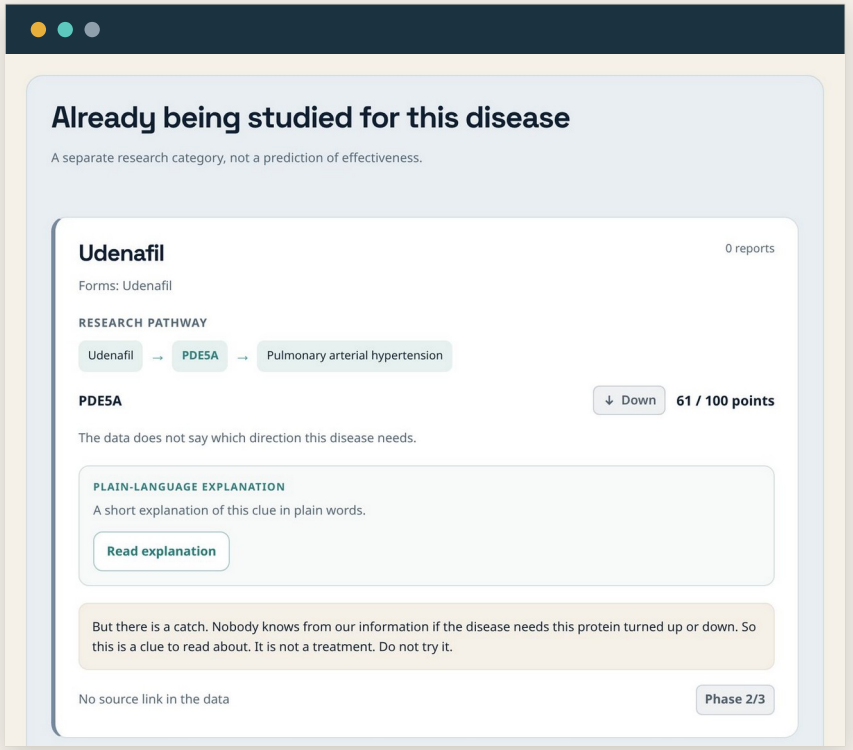
Medicines first: ranked clues, each with source, reports and direction

05 · EVERY CLUE SHOWS ITS WORKING

Source, strength and direction on every clue.



Source, evidence strength and direction on every clue



Medicines already in trials are shown separately, with their phase

Screenshots from a local run of the public code on live Open Targets data, 13 Nov 2026. Explanation shown uses the built-in template; Gemini writes these when the key is set.

06 · TECH STACK

Open data, Gemini, and 300+ automated tests.

Layer	Choice	Why
Data	Open Targets Platform GraphQL API	Free, open-licensed, maintained by EMBL-EBI and Sanger. Disease, target and drug records with evidence scores.
Engine	Python 3; deterministic ranking in plain code, no AI	Asks Open Targets twice and saves nothing if the answers disagree. Cache keyed by Open Targets data version (26.09). Parallel fetch.
AI	Gemini (gemini-3.5-flash) via REST	Explains only the retrieved records. Key comes from the environment, never in code or the image.
Interface	FastAPI + HTML page	Endpoints /api/search, /api/clues, /api/explain. Medicines shown first.
Hosting	Docker container ready for Google Cloud Run	Non-root, no key baked in. A public link is the next step.
Build process	GitHub + Claude Code; small reviewed tickets	300+ automated tests. Kashish merges only what matches her paper graph.

Secrets live in the environment, never in code. ChEMBL is an optional future backup source.

A lookup shows what is known. A detective removes it.

Open Targets already lists the drugs being tested for a disease. Those are not clues. Moonlight subtracts them, then ranks what remains by how many disease-linked proteins each medicine touches and how strong those links are.

```
candidates = drugs(targets(disease))   − known(disease)   → rank(score · hits)
```

08 · SAFETY BY DESIGN

Clues, never answers. The guardrails were written before the first line of code.

⚠ Every page: for research exploration, not medical advice. Never change a medicine without a doctor.

◉ Every result shows its source and evidence strength. No AI-invented links; Gemini explains only retrieved records.

⇅ Every result shows direction: does the medicine turn the protein up or down? The wrong direction can worsen a disease.

⌕ Honest gaps are stated: when the data does not say which direction the disease needs, the result says so.

Framed as “connections worth reading about”, never as treatments.

What we learned from a real rare disease.

What we found

The engine worked. But only 1 of the top 25 disease proteins had any medicine, and its one clue, ocriplasmin via LAMA2, may point the wrong way: it cuts the very protein patients lack.

What it missed

Four trial medicines linked directly to the disease were not shown:

Ribitol · Phase 3
Prednisone · Phase 2
A gene therapy · Phase 1/2
ATYR-1940 · Phase 1/2

The V1 fix

Our “already being studied” panel only reaches trial drugs through the top proteins. V1 widens it to medicines linked directly to the disease, with their phase (“being studied, not proven”), adds clearer wrong-direction warnings, and asks which subtype is meant.

An honesty and safety lesson: show what is already being tested, and flag clues that may point the wrong way.

Free for the curious. Paid for the organised.

FREE TIER

Students, teachers, patient families, curious adults

- Plain-language search and clickable map
- Sources, evidence strength and direction on every result
- Builds trust and reach; the underlying data is public anyway

ORGANISATION TIER · ANNUAL SUBSCRIPTION

Rare-disease foundations, advocacy groups, research teams, biotech analysts

- Saved disease watchlists and alerts on each database release
- Team workspaces
- Exportable, citation-backed briefing reports for researchers and grant panels
- Sponsored access for small patient groups

Primary market: India. Secondary: US and UK, home to the largest rare-disease foundation ecosystems.

We complement the research platforms. We do not compete with the scientists.

Alternative	Built for	Moonlight's difference
Open Targets Platform website	Scientists who know the ontology and the scores	Plain-language search, a map, and the known-drug filter on top of their data
AI repurposing programmes (e.g. Every Cure)	Researchers and clinicians, via large internal pipelines	Self-serve, public, explainable tool for non-specialists and advocacy groups
Academic knowledge-graph tools	Users with graph and query skills	No setup, no code; sources, direction and evidence strength on every result

Non-specialist first. Explanations a curious 12-year-old can follow.

Detective, not lookup. Known drugs removed, remaining clues ranked.

Safety first. AI explains retrieved data only; nothing invented.

The SMAKS Project: a family team from Kochi, two co-founders.

Shreshth Dugar

CO-FOUNDER · BUSINESS LEAD

CMO of a 100-year-old listed tea plantation company, COO of a bulk tea supplier, NYU-trained journalist. Leads the business and humanitarian case.

Kashish Dugar, 9

CO-FOUNDER · TECH LEAD

Codes 2.5 hours a day and blogs it. Finished Grokking Algorithms in 18 days, each algorithm built from scratch in Python. Deployed a Django app; now learning Rust. Designs the ranking on paper.

Sidha Mogha

DESIGNER · FOUNDER, COCHIN DESIGN COMPANY

Agastya Dugar

TESTER

Stage: working product (13 Nov)

STATUS

- Engine done (fetch, filter, rank, explain) with 300+ tests
- Web page and API merged
- Speed work merged
- Cloud Run container ready
- Next: public Cloud Run link, wider trials panel, everyday-language search
- Google credits fund Gemini usage and Cloud Run

Blog: kashishdugar2017.github.io · Repo: github.com/kashishdugar2017/moonlight · Contact: shreshthdugar@gmail.com